Practice workbook · Volumes of revolution between two boundaries

Finding volumes about the x-axis for regions between a curve and a line or two curves, by subtracting or adding solids.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-23

1. Question 2 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

2 The finite region enclosed by the curve with equation $y = 4 - x^2$ and the line with equation $y = x + 2$ is rotated through 360° about the x-axis.

Use algebraic integration to find the exact volume of the solid formed.

(6)

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1. Question 2 · part whole · 6 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 6 marks)

1. Question 2 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

10

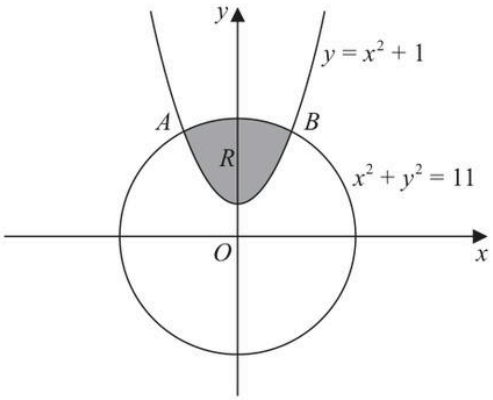


Diagram **NOT** accurately drawn

Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = x^2 + 1$ and the curve with equation $x^2 + y^2 = 11$

The two curves intersect at the point A and at the point B.

(a) Find the x coordinate of the point A and the x coordinate of the point B.

(4)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume, to 2 decimal places, of the solid generated.

(5)

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2. Question 10 · part b · 5 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

Question 10 continued

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2. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

Question 10 continued

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(Total for Question 10 is 9 marks)

2. Question 10 · part b · 5 marks · Working and answer

Working and answer

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3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

10 $f(x) = 32x^3 - 33x + 1$

(a) Show that $f(1) = 0$

(1)

(b) Hence using an algebraic method solve $f(x) = 0$

(4)

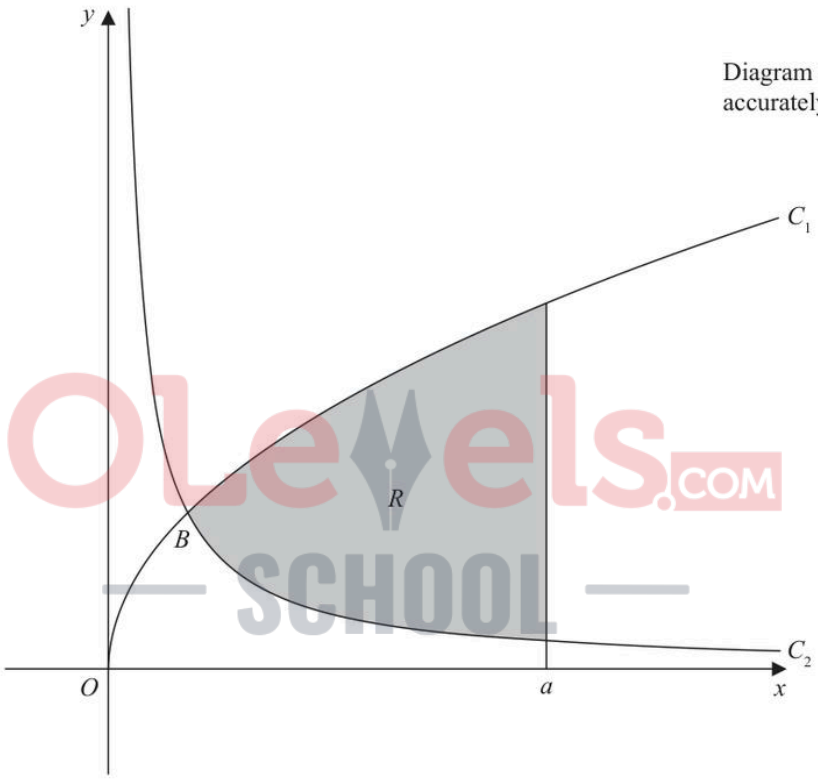


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve C_1 with equation $y = \sqrt{x}$, by the curve C_2 with equation $y = \frac{1}{8x}$ and by the line with equation $x = a$

The curves C_1 and C_2 intersect at the point B, with x coordinate p, where $p < a$

(c) Find the value of p.

(2)

The region R is rotated through 360° about the x-axis to generate a solid with volume $\frac{27\pi}{64}$

(d) Use algebraic integration to find the value of a.

(7)

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3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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3. Question 10 · part d · 7 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 10 continued

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(Total for Question 10 is 14 marks)

3. Question 10 · part d · 7 marks · Working and answer

Working and answer

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4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

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Diagram **NOT** accurately drawn

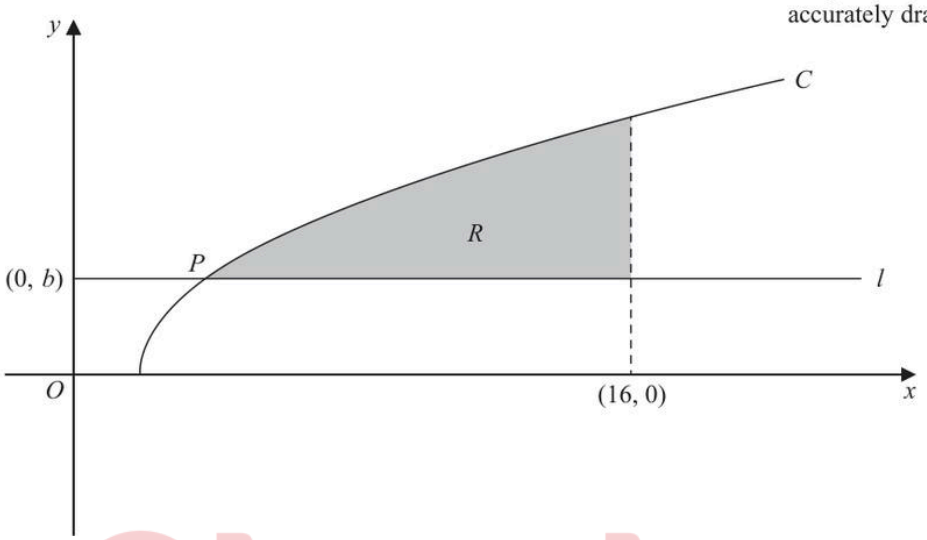


Figure 2

Figure 2 shows part of the curve C with equation $y = \sqrt{x - 2}$
Figure 2 also shows the straight line l with equation $y = b$ for $x > 0$ where $b > 0$

Given that C and l intersect at the point P with coordinates (a, b), where $2 < a < 16$

(a) show that $b^2 = a - 2$

(2)

The finite region R bounded by C, the straight line with equation $x = 16$ and l, shown shaded in Figure 2, is rotated through 360° about the x-axis to form a solid S.

Given that the volume of the solid formed is 50π

(b) use algebraic integration to find the value of a and the value of b.

(9)

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4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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4. Question 11 · part b · 9 marks

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Further practice · Attempt part b only.

Question 11 continued

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4. Question 11 · part b · 9 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 11 continued

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(Total for Question 11 is 11 marks)

4. Question 11 · part b · 9 marks · Working and answer

Working and answer

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5. Question 11 · part whole · 8 marks

2021 JUN · Paper 02

Further practice · Attempt part whole only.

11

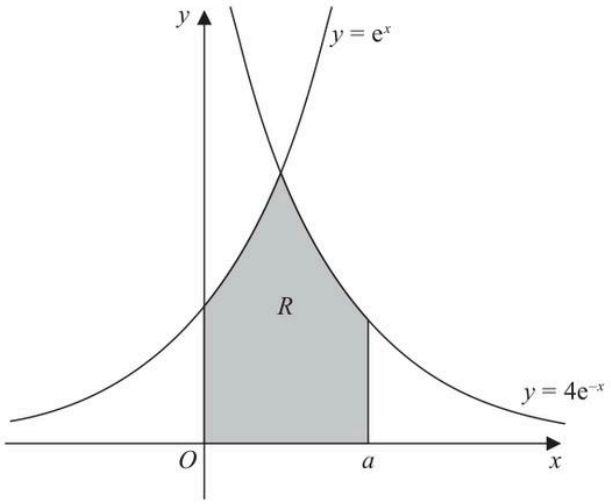


Diagram **NOT** accurately drawn

Figure 4

The region R, shown shaded in Figure 4, is bounded by the curve with equation $y = e^x$, the curve with equation $y = 4e^{-x}$, the straight line with equation $x = a$, the x-axis and the y-axis.

When the region R is rotated through 360° about the x-axis, the volume of the solid generated is

$$k - 8\pi e^{-4}$$

where k is a constant.

Using algebraic integration, find a possible value of a and the exact corresponding value of k.

(8)

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5. Question 11 · part whole · 8 marks

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Further practice · Attempt part whole only.

Question 11 continued

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5. Question 11 · part whole · 8 marks

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Further practice · Attempt part whole only.

Question 11 continued

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(Total for Question 11 is 8 marks)

5. Question 11 · part whole · 8 marks · Working and answer

Working and answer

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6. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

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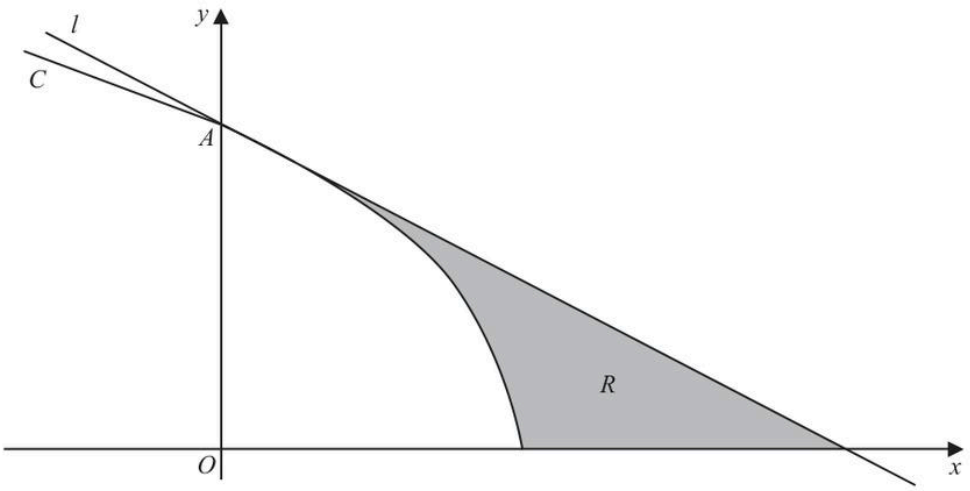


Figure 3

Figure 3 shows part of the curve C with equation $y = 3\sqrt{4-x}$ and part of the line l with equation $px + qy + r = 0$ where p and r are integers and q is prime.

The curve C cuts the y-axis at the point A

The line l is the tangent to C at A

(a) Find the value of p, the value of q and the value of r

(5)

The finite region R, shown shaded in Figure 3, is rotated through 360° about the x-axis.

(b) Use algebraic integration to find the volume of the solid formed.
Give your answer in terms of π

(5)

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6. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

Question 10 continued

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6. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

Question 10 continued

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6. Question 10 · part b · 5 marks

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Further practice · Attempt part b only.

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Question 10 continued

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(Total for Question 10 is 10 marks)

6. Question 10 · part b · 5 marks · Working and answer

Working and answer

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7. Question 9 · part whole · 10 marks

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Further practice · Attempt part whole only.

9

Diagram **NOT** accurately drawn

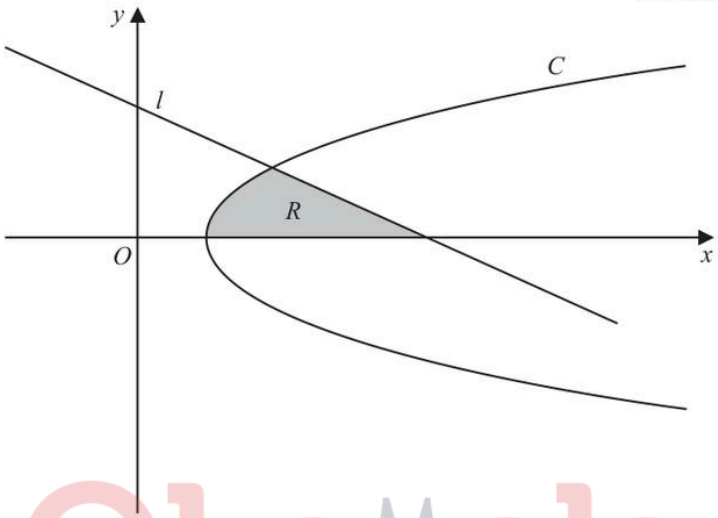


Figure 3

Figure 3 shows part of the curve C with equation $y^2 = x - 1$ and part of the line l with equation $2y + x - 4 = 0$

The region R, bounded by the x-axis, the curve C and the line l, is rotated through 360° about the x-axis.

Using algebraic integration, find the exact value of the volume of the solid generated.

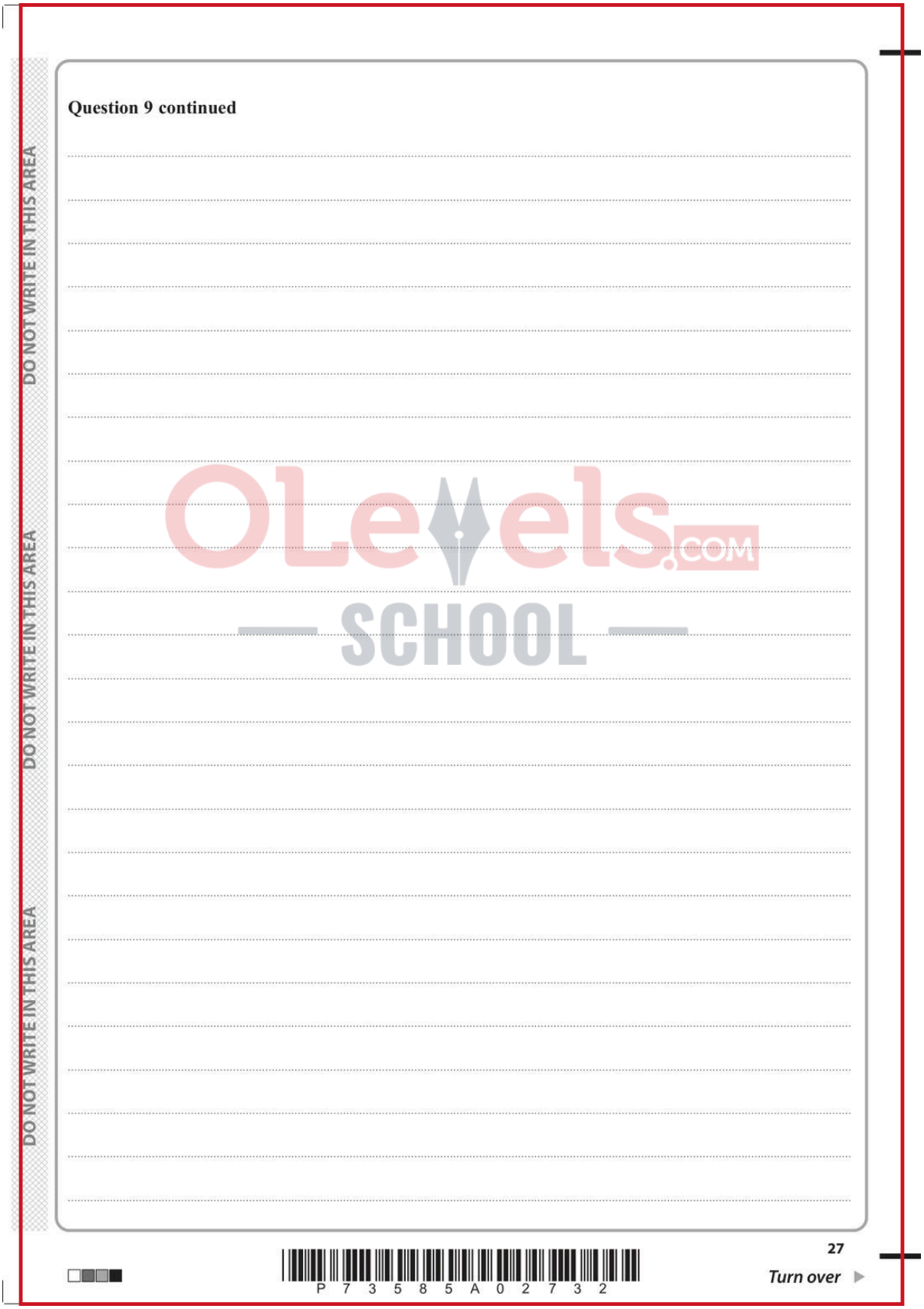
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7. Question 9 · part whole · 10 marks

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Further practice · Attempt part whole only.



Question 9 continued

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P 7 3 5 8 5 A 0 2 7 3 2

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Turn over

7. Question 9 · part whole · 10 marks

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Further practice · Attempt part whole only.

Question 9 continued

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7. Question 9 · part whole · 10 marks

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Further practice · Attempt part whole only.

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Question 9 continued

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(Total for Question 9 is 10 marks)

7. Question 9 · part whole · 10 marks · Working and answer

Working and answer

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8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

6

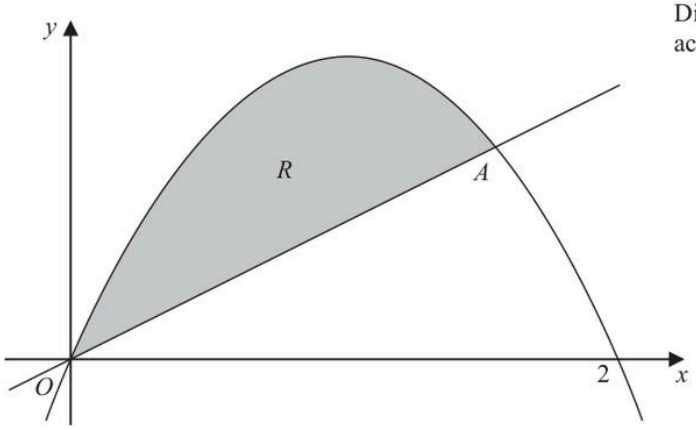


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Figure 2

The region R, shown shaded in Figure 2, is bounded by the curve with equation $y = 2x - x^2$ and the line with equation $2y - x = 0$

The curve and the line intersect at the origin O and the point A.

(a) Show that the point A has coordinates $\left(\frac{3}{2}, \frac{3}{4}\right)$.

(2)

The region R is rotated through 360° about the x-axis.

(b) Use algebraic integration to find, in terms of π, the volume of the solid formed.

(6)

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8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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8. Question 6 · part b · 6 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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8. Question 6 · part b · 6 marks

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Further practice · Attempt part b only.



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8. Question 6 · part b · 6 marks · Working and answer

Working and answer

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9. Question 5 · part whole · 8 marks

2023 NOV · Paper 01

Further practice · Attempt part whole only.

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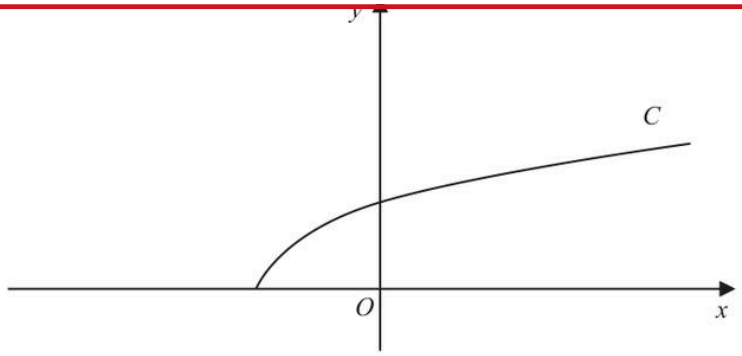


Figure 2

Figure 2 shows the graph of part of the curve C with equation $y = \sqrt{2x+6}$
The finite region enclosed by the curve C and the straight line with equation $3y - x = 3$ is rotated through 360° about the x-axis.

Use algebraic integration to find the exact volume of the solid generated.
Give your answer in terms of π

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9. Question 5 · part whole · 8 marks

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Further practice · Attempt part whole only.

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Question 5 continued

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(Total for Question 5 is 8 marks)

9. Question 5 · part whole · 8 marks · Working and answer

Working and answer

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10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

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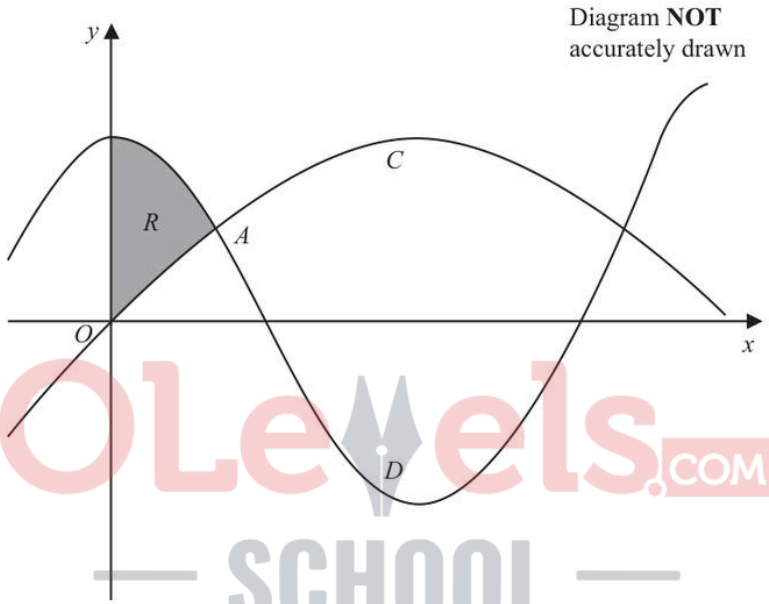


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

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10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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Question 10 continued

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10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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10. Question 10 · part c · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 10 continued

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(Total for Question 10 is 15 marks)

10. Question 10 · part c · 5 marks · Working and answer

Working and answer

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